



Facile fabrication of polyacrylate-siloxane antifouling coatings with self-generated biomimetic surface

Haofeng Xiao^a, Kai Zhang^b, Taijiang Gui^b, Shuxue Zhou^{a,*}

^a College of Smart Materials and Future Energy, State Key Laboratory of Advanced Coatings for Equipment, Advanced Coatings Research Center of Ministry of Education of China, Fudan University, Shanghai 200433, China

^b Marine Chemical Research Institute Co., Ltd., State Key Laboratory of Advanced Coatings for Equipment, Qingdao 266071, China

ARTICLE INFO

Keywords:

Antifouling coatings
Biomimetic Surface
Micro-nano structure
Antibacterial effect
Underwater superoleophobicity

ABSTRACT

Antifouling coatings with biomimetic micro-nano structure have been widely explored for the replacement of traditional self-polishing coatings that relied on high content of biocides. However, their application is limited by the complicated preparation process as well as the lack of antibacterial effect and durability of micro-nano structure. Herein, a bactericide-free antifouling coating with capability to self-generate biomimetic surface was prepared using trimethoxysilyl/triisopropylsilyl ester-bearing polyacrylate and amino-silane coupling agent. Owing to the spontaneous phase separation of amino-silane, a micro-nano textured surface with both underwater superoleophobicity and alkaline hydration layer was generated under seawater immersion, mimicking the anti-adhering and bactericidal functions of the skins of pilot whales and dolphins. Interestingly, identical self-polishing rate and stable biomimetic surface were exhibited both in static and dynamic immersion tests, resulting from the combined action of the micro-nano structure and the alkaline hydration layer. Although no bactericide was contained, the coating based on [3-(2-aminoethyl)aminopropyl]trimethoxysilane possessed sterilization rate of 97.3 % (*E. coli*) and anti-algae rate of 85.1 % (*Chlorella* sp.), and could maintain its sterilization rate at 93.1 % and 94.2 % even after 15 days of static or dynamic immersion, respectively. Meanwhile, the coating possessed superior antifouling performance and durability over conventional biomimetic surfaces in the actual marine environment, which came from the combination of micro-nano structure with intrinsic alkalinity and self-regeneration ability. Overall, our work would provide a new and practical strategy for the development of biomimetic antifouling coatings.

1. Introduction

With the prosperity of global trade and the exploit of marine resources, biofoulings are causing non-negligible losses to the marine economy, as a large number of fouling organisms will lead to an increase in energy consumption as well as maintenance cost of ships and related facilities. The application of antifouling coatings (AFCs) provides an effective strategy to address this problem by inhibiting the attachment of biofoulings or directly killing the fouling organisms already attached to the surface [1]. According to antifouling mechanism, antifouling coatings can be divided into three main categories, that is, fouling-degrading coatings, fouling-resistant coatings and fouling-release coatings [2–6].

So far, fouling-degrading coatings based on various self-polishing copolymers dominate commercial AFCs market, due to their low cost, reliable antifouling performance and high durability. Typical self-

polishing antifouling coatings are generally composed of copolymer bearing hydrolyzable side groups and antifoulants such as Cu₂O and 4,5-dichloro-2-n-octyl-4-isothiazolin-3-one [7–9]. Despite excellent antifouling performance, traditional self-polishing antifouling coatings have a series of drawbacks, including high content of toxic biocides, uncontrolled release of antifoulants and poor static antifouling performance [10]. Therefore, varied strategies have been reported for antifouling coatings. The most common approach is to synthesize intrinsic marine antifouling resin by covalently binding antibacterial groups to polymer chains, such as quaternary ammonium salt [11], capsaicin [12], eugenol [13] and isoborneol [14]. Another effective way is to introduce a degradable backbone or crosslinking agent to control the release rate of antifoulants and enhance their antifouling efficiency [15,16]. Xu et al. [17] synthesized polyurethane with poly(ϵ -caprolactone) segments and triisopropylsilyl ester pendant groups. Due to the enzymatic degradation

* Corresponding author.

E-mail address: zhoushuxue@fudan.edu.cn (S. Zhou).

<https://doi.org/10.1016/j.cej.2025.164087>

Received 26 February 2025; Received in revised form 25 April 2025; Accepted 22 May 2025

Available online 23 May 2025

1385-8947/© 2025 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

of the poly(ϵ -caprolactone) main chain and the rapid hydrolysis of triisopropylsilyl ester groups, the coating could maintain a sufficient self-polishing rate even in static marine environments. Nevertheless, these strategies still cannot get rid of the dependence on biocides, as antifoulant-free coatings with decent antifouling performance have not been reported yet.

Since the nature has evolved numerous strategies to deal with fouling adhesion, the combination with biomimetic principle has also become an emerging method to fabricate environmentally friendly antifouling coatings [18]. Based on the bioinspiration, a large number of antifouling coatings with biomimetic micro-nano topography have been developed. By mimicking the physical structures on the skin surface of marine organisms, these coatings can effectively prevent fouling adhesion through attachment point reduction [19,20]. Liu et al. [21] utilized polydimethylsiloxane (PDMS) containing nano-silica to fabricate biomimetic shark skin surface with lotus effect through the microreplication method and flame treatment. Brzozowska et al. [22] prepared micropatterned PDMS surface that mimicked the tubercle and spine structures on the shell surface of *M. hardwickii* by lithography. Cao et al. [23] prepared a polyelectrolyte multilayer coating composed of poly(acrylic acid) and polyethyleneimine through layer-by-layer methods. The obtained coating surface was featured by micro-scaled ridges and pores that were similar to the morphology of the pilot whale skin. Since most micro-nano structures are constructed by a relatively complex preparation process, these biomimetic surfaces are still unable to achieve large-scale preparation and application [24–26].

On the other hand, some studies pointed out that coatings relying on biomimetic structures have difficulty in dealing with different biofoulings in the complex marine environment. Antibacterial groups have to be incorporated to obtain adequate service performance [22,27]. Wang et al. [28] introduced quaternary ammonium-functionalized SiO₂ nanoparticles into the self-polishing polymer matrix to achieve the combination of biomimetic micro-nano structured surface and antibacterial function. Nevertheless, the micro-nano structure in the coatings contributed less to the antifouling performance except for low oil adhesion. Zhao et al. [29] prepared a PDMS-based biomimetic coating with the microstructure of *Laminaria japonica* and the surface was further modified with layers of guanidine-hexamethylenediamine-PEI/sodium alginate and capsaicin containing nanocapsules through layer-by-layer self-assembly methods. The biomimetic surface could effectively reduce the adhesion of fouling organisms, and the release of capsaicin from nanocapsules enhanced the antifouling performance of the coating by inhibiting the reproduction of bacteria. Despite the progress made in antifouling efficiency for coatings based on biomimetic micro-nano structures, their practical application still suffers from the lack of long-term antifouling ability. This is because the antifouling performance of these coatings is overdependent on the topography and chemical composition of the initial surface, while the coating matrix itself lacks antifouling ability. Once the micro-nano structures are damaged by mechanical wear or covered by fouling organisms, the coating surface will present an irreversible reduction in antifouling performance [30,31]. So far, few studies have achieved the facile preparation of biomimetic coatings and their antifouling ability in the actual marine environment without the use of antifoulants.

Targeting at the demand of environmentally friendly antifouling coatings and overcoming the drawbacks of conventional biomimetic surfaces, we fabricated a polyacrylate-siloxane antifouling coating with self-generated biomimetic surface by a facile and cost-effective method. The coating consisted of only two components, i.e., trimethoxysilyl/triisopropylsilyl ester-bearing polyacrylate resin and amino-silane coupling agent. With polyamino-silane coupling agent, the coating could spontaneously separate into a dispersed phase enriched with amino-siloxane and a continuous phase enriched with polyacrylate during the curing process. Due to the difference in dissolution rate between the two phases, a biomimetic micro-nano textured surface with underwater superoleophobicity and highly alkaline hydration layer

could be generated under seawater immersion. By adjusting the monomer composition of the resin, the coating surface could also exhibit a stable self-polishing rate and maintain underwater superoleophobicity under both static and dynamic conditions. Through in-vitro bacteriostatic test, laboratory antifouling assessment and marine field test, it was confirmed that the self-generated biomimetic antifouling surface could achieve long-lasting bactericidal effect, excellent anti-protein and anti-algae ability, and good antifouling performance in actual marine environment, without the need for biocides.

2. Experimental section

2.1. Materials

Butyl acetate (chemical grade), tetrahydrofuran (THF, chemical grade), methyl methacrylate (MMA, chemical grade), butyl acrylate (BA, chemical grade), styrene (St, chemical grade) and special indicator paper (pH 6.5–10.0) were purchased from Sinopharm Chemical Reagent Co., Ltd. 3-(Trimethoxysilyl)propyl methacrylate (TMSPM, 99 %) was purchased from Shanghai Macklin Biochemical Technology Co., Ltd. Triisopropylsilyl methacrylate (TISM, 95 %) was purchased from Wuhan Huaxiang Kejie Biotechnology Co., Ltd. γ -Aminopropyl triethoxysilane (N1, 99 %), [3-(2-aminoethyl)aminopropyl]trimethoxysilane (N2, 95 %), [3-(2-aminoethyl)aminopropyl]dimethoxymethylsilane (N2_{DMM}, 95 %), [3-(2-aminoethyl)aminopropyl]triethoxysilane (N2_{TE}, 95 %) and 3-[2-(2-aminoethylamino)ethylamino]propyl trimethoxysilane (N3, 95 %), were purchased from Aladdin Chemical Reagent (China). 1,1-Di(*tert*-amylperoxyl)cyclohexane initiator (concentration: 80 %) was purchased from ARKEMA (Changshu) Chemical Co., Ltd. Fluorescein isothiocyanate-labelled bovine serum albumin (FITC-BSA, concentration: 5 mg·mL⁻¹) was purchased from Beijing Solarbio Science & Technology Co., Ltd. *Escherichia coli* (*E. coli*) and *Shewanella loihica* (*S. loihica*) were provided by Shanghai Microbiological Culture Collection Co., Ltd. *Chlorella* sp. and *Cyclotella cryptica* suspension (concentration: 10⁶ cfu·mL⁻¹) were obtained from Shanghai Guangyu Biological Technology Co., Ltd. Artificial seawater (ASW, pH = 8.2) was prepared in our lab according to ASTM D1141-98. Phosphate buffer saline (PBS, pH = 7.4) was prepared according to GB/T 20944.3–2008. All reagents were used as received.

2.2. Synthesis of trimethoxysilyl/triisopropylsilyl ester-bearing polyacrylate

Polyacrylate-siloxane was synthesized through free-radical solution polymerization. A four-neck flask with a mechanical stirrer was charged with butyl acetate and heated to 127 °C. After the solvent began to reflux, a certain amount of MMA, BA, St, TMSPM, TISM and initiator, that were mixed in advance, were added to the flask dropwise. The reactants were kept at 127 °C for 5 h, after which the reaction temperature was raised and kept at 132 °C for another half hour. Thereafter, trimethoxysilyl/triisopropylsilyl ester-bearing polyacrylate solution was obtained after cooling down to room temperature. The recipe and molecular weight of different resins were listed in Table 1. Since the polymerization conversion of all resins was beyond 98.5 %, the as-obtained resin solution was directly used in the preparation of coatings.

2.3. Preparation of polyacrylate-siloxane antifouling coatings

The as-obtained trimethoxysilyl/triisopropylsilyl ester-bearing polyacrylate solution, 4 g, was first diluted with 1 g butyl acetate, and then 0.28 g amino-silane coupling agent (namely N1, N2, or N3) were added as a curing agent. After being magnetically stirred for 20 min, the coating was cast on glass slides (7.5 × 2.5 cm²) using a 120 μ m draw-down rod. The coatings were then cured at room temperature for at least one week. The formulation of coatings was listed in Table 2.

Table 1

Mass fraction of monomers and polymerization results of trimethoxysilyl/triisopropylsilyl ester-bearing polyacrylate.*

Name	MMA/%	BA/%	St/%	TISM/%	TMSPM/%	Initiator/%	Conversion/%	M_n	PDI
T21	25.9	25.9	20.7	20.7	0	6.8	98.7	20,391	1.6
K21	25.9	25.9	20.7	0	20.7	6.8	99.3	25,231	1.8
T7K21	22.5	22.4	20.7	6.9	20.7	6.8	98.9	18,310	1.7
T21K7	22.5	22.4	20.7	20.7	6.9	6.8	98.6	16,403	1.8
T14K14	22.5	22.4	20.7	13.8	13.8	6.8	99.1	19,810	1.7

* The as-obtained trimethoxysilyl/triisopropylsilyl ester-bearing polyacrylate solution contains 30% of butyl acetate by mass fraction.

Table 2

The recipe of polyacrylate-siloxane coatings.

Sample name	Trimethoxysilyl/triisopropylsilyl ester-bearing polyacrylate solution, g	Butyl acetate, g	N1, g	N2, g	N3, g
T14K14-N1	4.00	1.00	0.28	0	0
T14K14-N2	4.00	1.00	0	0.28	0
T14K14-N3	4.00	1.00	0	0	0.28
T21-N2	4.00	1.00	0	0.28	0
K21-N2	4.00	1.00	0	0.28	0
T21K7-N2	4.00	1.00	0	0.28	0
T7K21-N2	4.00	1.00	0	0.28	0

2.4. Characterization

The dried resin and coating samples were characterized by a Nicolet IS5 Fourier-transform infrared spectrometer (FT-IR, Thermo Fisher Scientific, USA) in attenuated total reflection (ATR) mode. The gel fractions (GF) of coatings were measured by immersing the sample in THF for 2 days and calculating according to the equation W_{redry}/W_d , where W_d and W_{redry} are the initial weight of coatings and the re-dried weight of the swelled coating, respectively. Three replicates for each coating were adopted in the measurement. According to GB/T 6739–2022 and GB/T 9286–2021, the hardness and adhesion grade of coating samples on the steel substrate were obtained via a pencil hardness tester (Type 101 pencil, China First Pencil Co., Ltd) and a cross-cut tester, respectively. The chemical composition of coating surfaces was measured by a Thermo Fisher Nexsa X-ray photoelectron spectroscopy (XPS, Thermo Fisher Scientific, USA). The surface morphologies and cross sections of coatings were observed by a Gemini SEM500 field-emission scanning electron microscope (FE-SEM, Zeiss, Germany) at accelerating voltage of 3 kV. All coating samples were sputter-coated with gold prior to observation. The surface topography and elastic modulus distribution of coatings were probed by a Bruker Dimension Icon atomic force microscope (AFM, Bruker, USA) with a probed area of $2 \times 2 \mu\text{m}^2$. The thickness of coatings was observed by a KH-7700 digital microscope (Hirox, Japan). The water contact angles (WCA) and underwater oil contact angles (UWOCA) were measured using an OCA25 contact angle analyzer (Dataphysics, Germany). The average values of WCA and UWOCA were adopted from at least five parallel measurements using a 2 μL deionized water (DIW) droplet and a 4 μL hexadecane droplet, respectively. The pH value of the solution was obtained by a PH5F flat pH tester (Sanxin Apera, China). The surface Zeta potential of coating was characterized by a SurPass3 electrokinetic analyzer (Anton Paar, Austria) under the pH of 8.2.

2.5. Testing of static and dynamic immersion

In static immersion test, the coating samples were directly submerged in 300 mL of ASW and taken out at set intervals to track their UWOCA. A rotating device shown in Fig. S1 was used for dynamic

immersion test. The coating samples were fixed in the grooves of the device and subsequently scoured in ASW at a speed of 400 rpm to simulate 10 knots of the ship's navigational speed. The samples were rotated in ASW for 15 days and taken out every 3 days to check the integrity of the coatings and track their UWOCA.

The water absorption ratio and mass loss of coatings were also measured through immersion tests. The weights of glass substrates and cured samples were recorded as W_g and W_0 , prior to testing. When the static and dynamic immersion tests were completed, the coating samples were taken out and slightly rinsed with deionized water. After carefully removing water from the surface using filter paper, the weight of wet sample was recorded as W_1 . The coating sample was then dried in an 80 °C oven to a constant weight, which was recorded as W_2 . The water absorption ratio was calculated as $(W_1 - W_2)/(W_1 - W_g)$ and the mass loss as $(W_0 - W_2)/A$, where A represents the coating area.

2.6. In-vitro bacteriostatic test

E. coli and *S. loihica* were chosen as target bacteria to determine the bacterial sterilization ability of polyacrylate-siloxane coatings. Before the test, all coating samples were exposed under UV irradiation for 1 h to eliminate the interference from residual bacteria on the surface. A 50 μL suspension of bacteria in nutrient broth, which had successfully undergone recovery, was dropped onto the coating surface and then covered with polyethylene film. After incubating at 37 °C with relative humidity over 90 % for 24 h, the biofilm was carefully washed over using 1 mL PBS eluent in total. The obtained bacterial stock solution was diluted to an appropriate concentration and then inoculated into nutrient Agar medium. After 24 h of culture at 37 °C, each petri dish was photographed to count the number of recovered bacteria, which was recorded as N_1 . The glass substrate was used as the blank control, whose number of recovered bacteria by the same method was recorded as N_0 . The sterilizing rate was obtained by $(N_0 - N_1)/N_0$, with three replicates for each test.

2.7. Laboratory antifouling assessment

Protein adhesion assays and algal adhesion assays were carried out to examine the laboratory antifouling performance of coatings. FITC-BSA was used as a model protein foulant for protein adhesion assays, and first diluted to a 20 $\mu\text{g/mL}$ solution with PBS prior to the test. The coating samples were then placed in the petri dishes containing 10 mL FITC-BSA solution and incubated at 25 °C for 7 h in dark. After that, the samples were taken out and slightly rinsed with PBS to remove the remaining free proteins. The quantity of proteins adhering to the coating surface was acquired by the fluorescence intensity observed by a confocal laser scanning microscopy (CLSM, Zeiss, Germany) under 488 nm excitation laser. Algal adhesion assays were conducted similarly, using *Chlorella* sp. and *Cyclotella cryptica* as model algae foulants. 10 mL of algae suspension were added to each petri dish after being sufficiently shaken. The coating samples were then placed in petri dishes and incubated at 25 °C for 24 h under light. After that, the samples were taken out and slightly rinsed with ASW to remove the remaining free algae. CLSM images of different samples were recorded under 405 nm excitation laser. For all the above experiments, the glass substrate was

used as the blank control.

2.8. Marine field test

The marine field tests were conducted at the offshore area of Yellow Sea, Pacific Ocean (Jiaozhou Bay, coordinates: 120°31' E, 36°08' N) during the booming season of marine organisms (from June 2024 to August 2024). The coatings were sprayed onto steel panels which were pre-coated with epoxy primer, and tested under both static and dynamic conditions. The static test lasted for 15 d by directly submerging the panels at approximately 1 m in depth. The panels in dynamic test were mounted on a rotating device (Fig. S1) and rotated underwater at a speed equivalent to about 20 knots of the ship's navigation speed. After terminating the submersion, the panels were taken out and photographed without washing. The steel panels only coated with epoxy primer were used as the blank control.

3. Results and discussion

3.1. Fabrication and basic properties of polyacrylate-siloxane coatings

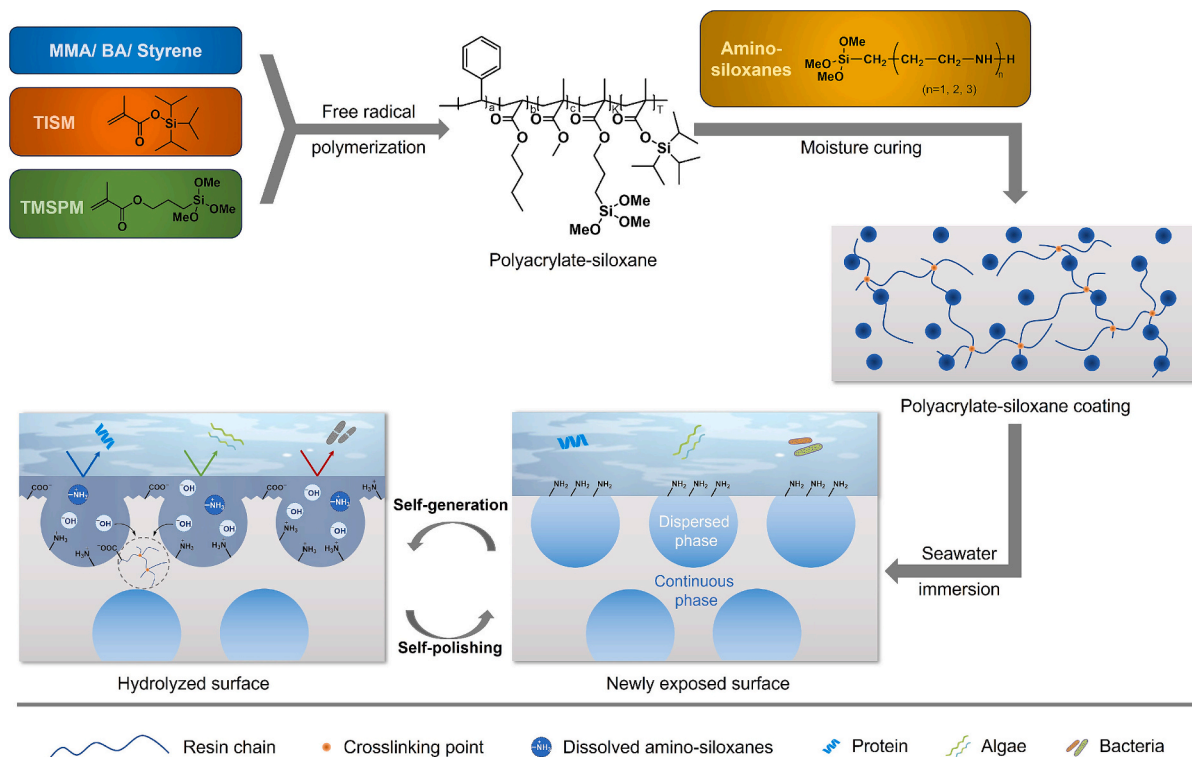
As shown in Scheme 1, TISM and TMSPM were chosen as the self-polishing monomer and the cross-linking monomer to synthesize trimethoxysilyl/triisopropylsilyl ester-bearing polyacrylate, respectively. By controlling the amount of initiator during radical polymerization, the obtained polymer possessed a relatively low average molecular weight (Table 1). The FT-IR spectra of the synthesized polyacrylate are presented in Fig. S2. The peaks at 2947, 1726, 1082 and 883 cm^{-1} were assigned to C-H, C=O, Si-O and Si-C bonds, respectively [32]. Due to the different chemical structure of TISM and TMSPM, their content in the resin could be reflected by the peak strength of Si-C and Si-O bonds. The results suggested successful synthesis of trimethoxysilyl/triisopropylsilyl ester-bearing polyacrylate.

The resins were then blended with amino-silane coupling agents to fabricate coatings. Amino-silane coupling agents are hygroscopic and

can catalyze the hydrolysis and condensation of siloxane groups, making trimethoxysilyl/triisopropylsilyl ester-bearing polyacrylate moisture-curable at room temperature (Fig. S3). The gel fraction, pencil hardness, and adhesive force were measured for the cured coatings, as shown in Fig. 1 and Fig. S4. It could be found that the number of amino groups contained in amino-silane coupling agent had little influence on the crosslinking degree and adhesion grade of the coatings. However, the pencil hardness of the coatings decreased roughly linearly with the increased number of amino groups, which might be related to the tendency of longer amino alkyl chains to clump together and form a flexible region. As for coatings with different monomer compositions, the gel fractions and mechanical properties were mainly affected by the content of TMSPM in resins. Due to the lack of siloxane groups in the resin, T21-N2 possessed lower adhesion grade than other coatings. In addition, T21-N2 and T21K7-N2 exhibited significantly lower crosslinking degree, and hence lower pencil hardness. As a whole, most of polyacrylate-siloxane coatings have satisfied pencil hardness (as high as 2H) and adhesive force (the best 0 grade), which matched the basic properties of coatings for practical application.

3.2. Self-generation of biomimetic antifouling surface

The surface topography and modulus distribution of coating with different amino-silane coupling agents were presented in Fig. 2a-c. It could be observed that the coating cured by N1 possessed a very smooth surface and the modulus of different regions of the surface showed no significant difference, indicating that a homogeneous mixing was achieved between the resin and N1 curing agent. However, when the amino-silane contained more than one amino group (i.e., N2 and N3), ordered pitting structures could be found on the surface of T14K14-N2 and T14K14-N3. The average radius of the pit structures was about 100 nm, and the modulus at the bottom of the pits was lower than the flat area on top. It should be noted that the density of pits on the surface of T14K14-N3 was greater than that of T14K14-N2, while the modulus at the bottom of pits decreased compared to T14K14-N2. The amino-silane N2 was



Scheme 1. Schematic diagram of the preparation of polyacrylate-siloxane coatings and the dynamic compositional changes of the coatings under seawater immersion.

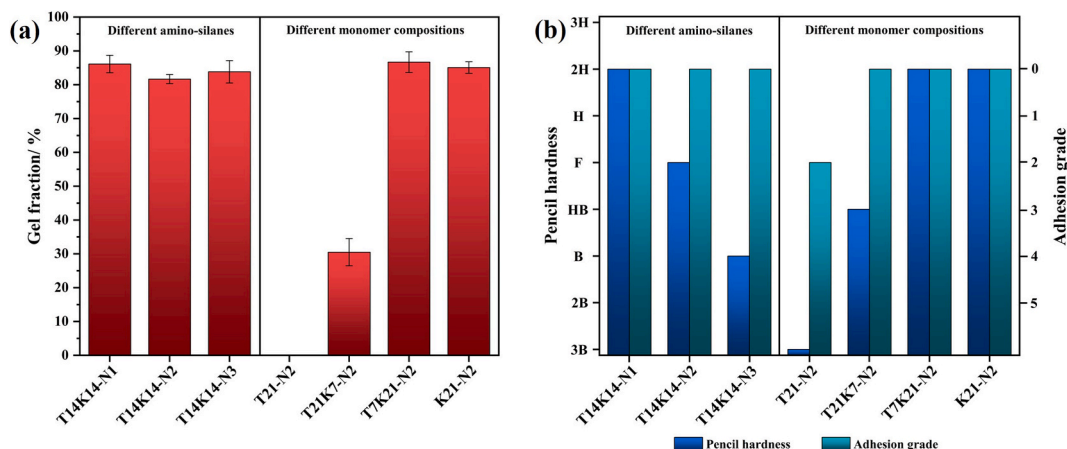


Fig. 1. (a) Gel fractions, (b) pencil hardness and adhesion grade of coatings. (Mean $\pm$ SE, $n = 3$).

also replaced by N_{2DMM} and N_{2TE} to fabricate the cross-linked polyacrylate-siloxane coatings, T14K14-N_{2DMM} and T14K14-N_{2TE}, respectively. Micro-nano porous structures were also formed on the surfaces of both coatings after 3 d of ASW immersion, as revealed in Fig. S5. Therefore, the formation of the pitting structures on coating surface might be related to the spontaneous phase separation of polyamino-silane coupling agents. As the solvent evaporated during the curing process of the coating, the long amino alkyl chains clumped together via the hydrogen bonding between each other, and the self-hydrolysis/condensation of amino-silanes further facilitated the phase separation. As a result, the coating spontaneously separated into a dispersed phase enriched with amino-siloxane and a continuous phase enriched with polyacrylate. Due to the difference in surface energy and density between the two phases, the dispersed phase on the coating surface eventually exhibited in the form of ordered pitting structures.

As shown in Fig. 2d-f, the spontaneous phase separation of amino-silane coupling agents further led to the formation of micro-nano structures on the surfaces of the coatings after immersion in ASW. T14K14-N1 maintained a smooth surface during the whole immersion process, as N1 was uniformly cross-linked with polyacrylate. Since the hydrophilicity of the dispersed phase was significantly higher than that of the continuous phase in T14K14-N2 and T14K14-N3, their surfaces presented inconsistent hydrolysis and dissolution rates in different regions, which led to the formation of biomimetic surfaces featured by micro-scaled pores and nano-scaled pits. It could be observed that, within the first hour of ASW immersion, T14K14-N2 and T14K14-N3 already exhibited porous structure on the surfaces and the arrangement of pores was consistent with the distribution of the original dispersed phase, verifying that the amino-silane coupling agent enriched dispersed phase was quickly hydrolyzed and dissolved during immersion. After the immersion was extended to 1 d, the pores on the surfaces exhibited an increased radius but a decreased number, which might be related to the hydrolysis of polyacrylate, making the coating surface begin to self-polish. The micro-nano structures no longer showed obvious changes from day 1 to day 3 of immersion, which indicated that the coatings could maintain stable surface morphology through continuous self-polishing. Among three coatings, T14K14-N2 presented the most porous surface with R_q of 87.6 nm after immersion (Fig. S6, determined by AFM characterization), suggesting the different hydrolysis process and self-polishing rates of coatings with different amino-silanes.

To better understand the changes in surface composition of coatings during the formation of micro-nano structures, FT-IR spectra of T14K14-N2 were recorded before and after immersion. As shown in Fig. 3a, the characteristic peaks at around 3300 and 1030 cm⁻¹ could be clearly found on the surface of T14K14-N2 before immersion, corresponding to the N-H bonds in the amino-silane coupling agent and Si-O-Si bonds in

the siloxane-crosslinked networks. When the coating was immersed in ASW for three days, the intensity of N-H related peaks decreased due to the dissolution of the domains enriched with amino-silane coupling agent. Meanwhile, the original sharp peak of Si-O-Si bonds transformed into a series of weaker peaks ranging from 990 to 1070 cm⁻¹. The appearance of these peaks might be related to the hydrolysis of cross-linking networks, exposing Si-O-H bonds on the coating surface. The FT-IR results confirmed that both amino-siloxane and polyacrylate matrix were involved in the hydrolysis process of the coating surface under ASW immersion (Fig. S7). Full XPS spectra of coatings were presented in Fig. S8, proving the existence of C, N, O, Si and Na elements on the surfaces after immersion. High-resolution XPS spectra of Na_{1s} and N_{1s} of different samples were also recorded for further analysis. As a result of the seawater-induced hydrolysis of triisopropylsilyl ester groups, the intensity of Na_{1s} peak reflected the content of the hydrolyzed polyacrylate chains on the coating surfaces [7]. The intensity of N_{1s} peak, on the other hand, reflected the content of amino-siloxane remained on the surfaces. As presented in Fig. 3b, almost no Na element existed on the surface of all coatings before immersion. After immersion in ASW for 3 days, the content of Na element on the coating surface showed varying degrees of increase, and the intensity of Na_{1s} peak decreased with more amino groups in amino-silane coupling agents. Given the fact that the content of self-polishing monomers was consistent in all coatings, this result might be explained by the difference in hydrophilicity and alkalinity of amino-silane coupling agents, which affected the hydrolysis of crosslinking networks and further resulted in different dissolution rates of the hydrolyzed species from the coating surface. As shown in Fig. 3c, T14K14-N3 possessed the highest content of N element on the surface before immersion, for the highest number of amino groups contained in N3. However, T14K14-N3 also exhibited the most significant decrease in the intensity of N_{1s} peak on the surface after immersion, indicating that the dissolution rate of N3 from the coating surface was the fastest and meanwhile much higher than the self-polishing rate of the polymer chain. Therefore, it could be inferred that T14K14-N3 possessed the fastest self-polishing rate under ASW immersion, with both amino-siloxanes and polymer chains rapidly dissolved from its surface, which also explained the relatively low roughness of self-generated micro-nano structures on its surface mentioned above.

Since the changes in surface morphology and surface composition of coatings under ASW immersion would cause changes in the surface wettability, which thereby affected the fouling-resistant ability, the UWOCAs and WCAs of coatings were measured as presented in Fig. 3d, e. For all three coatings, their UWOCAs increased during submersion, indicating the formation of hydration layers on their surfaces [33]. T14K14-N3 could achieve a leap in UWOCAs within the first hour of immersion, while T14K14-N1 and T14K14-N2 required at least one day of immersion, which was in agreement with its highest self-polishing

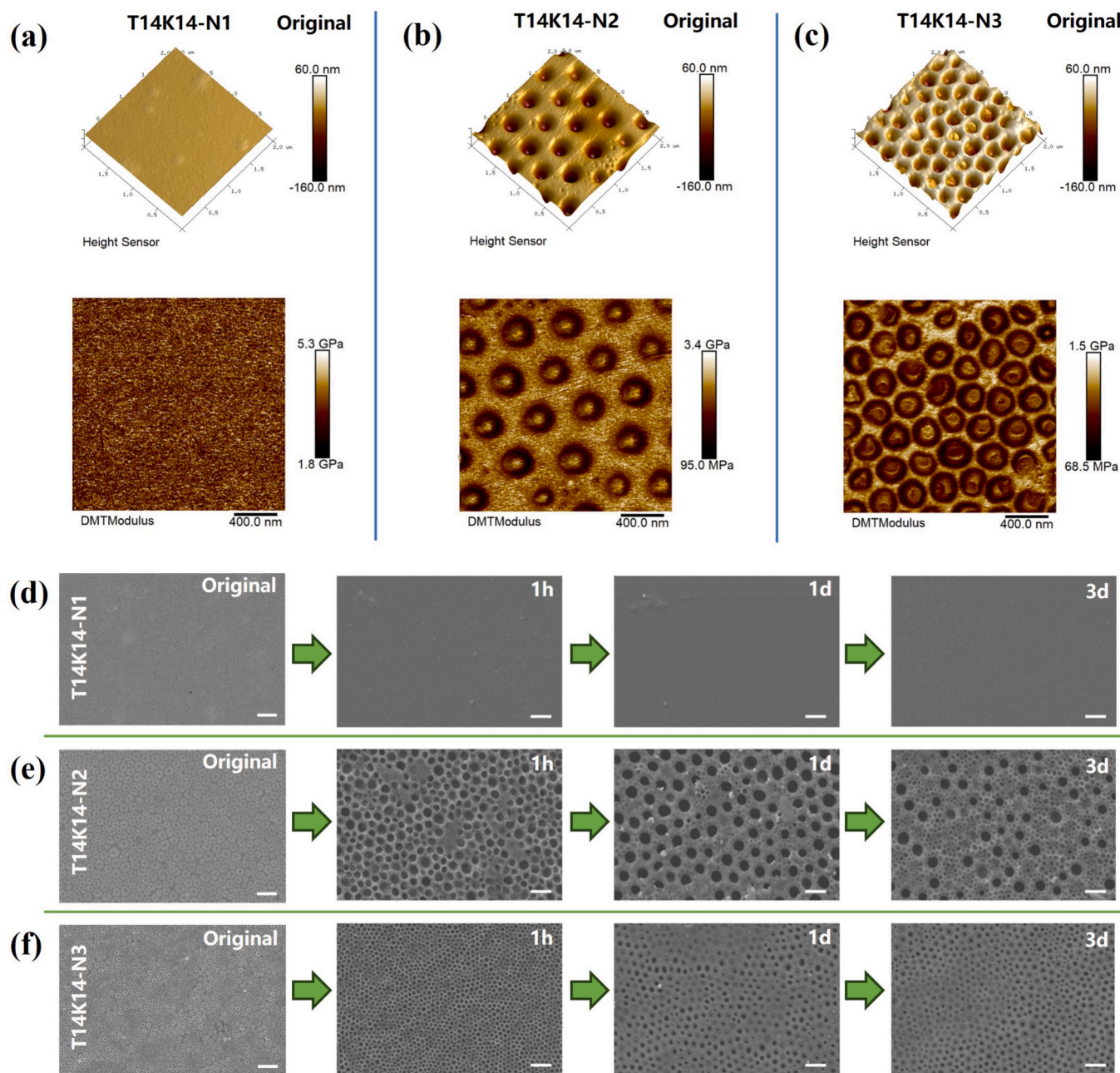


Fig. 2. AFM surface roughness and modulus distribution (a-c) of coating samples and SEM images of surfaces after immersion in ASW for different time (d-f). (SEM scale bar = 1 μm).

rate. It should be noted that, although the UWOCAs of all coatings significantly increased after immersion, only T14K14-N2 achieved an underwater superoleophobic surface (Fig. 3d and Fig. S9), which might result from the combination of the highly hydrophilic hydration layer and the coarsest micro-nano structures on its surface. As the hydration layer enhanced the wettability of water droplets on the surface, the WCAs of all coatings decreased just after immersion. Among them, T14K14-N2 presented the lowest WCA ($\text{WCA} = 56.3^\circ$), suggesting the highest degree of hydration on the surface. When the coatings were completely dried, the WCAs of T14K14-N2 and T14K14-N3 significantly increased and became even higher than those of the original surface, which was obviously related to the formation of surface micro-nano structures. Consistent with the SEM results, T14K14-N2 exhibited the highest hydrophobicity ($\text{WCA} = 120.2^\circ$). The surface alkalinity of coatings was qualitatively analyzed. That is, 200 μL of DIW was dropped

onto the coating surface, and the pH value of the droplet was measured using pH indicator paper after 10 min of wetting. As shown in Fig. 3f, the surface alkalinity of coatings increased with the number of amino groups in amino-silane coupling agents. Both T14K14-N2 and T14K14-N3 exhibited an obvious increase in the pH value of droplets as a result of the rapid dissolution of amino-siloxane enriched dispersed phase, while the droplet on the surface of T14K14-N1 remained neutral. The alkaline surface would provide an uninhabitable environment for fouling organisms.

3.3. Testing of static and dynamic immersion

Conventional biomimetic micro-nano topographical surfaces mostly lack durable antifouling ability, due to the loss of micro-nano structure by mechanical wear or covering of fouling organisms [34]. By virtue of

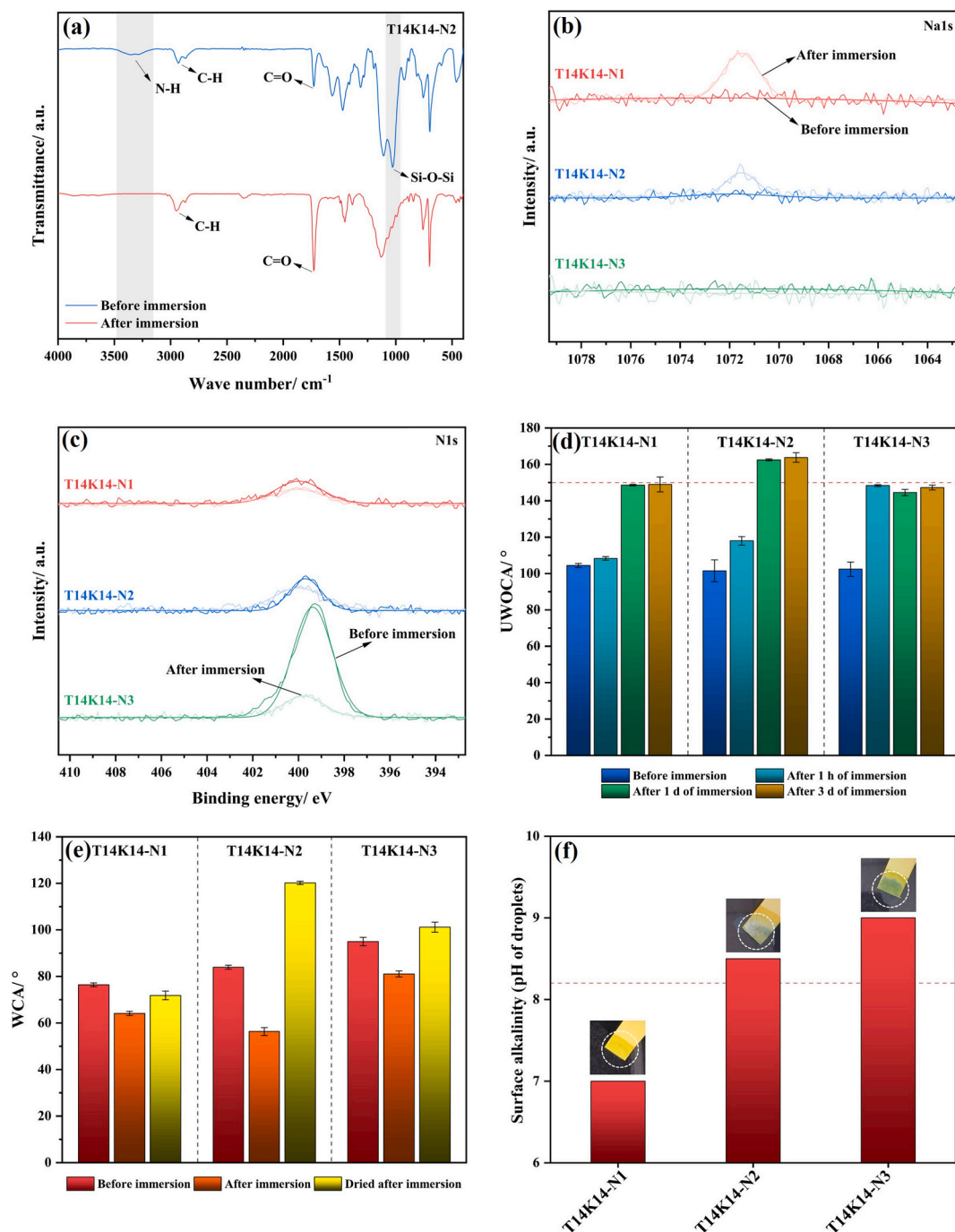


Fig. 3. IR spectrum (a) of T14K14-N2 and high-resolution XPS spectra of Na_{1s} (b) and N_{1s} (c) of coating samples before/after immersion; UWOCAs (d) and WCAs (e) of coating samples during immersion and the alkalinity (f) of the hydration layer on the surfaces. (Mean $\pm$ SE, $n = 3$; the quantitative data of XPS tests are listed in Table S1.).

the self-generated micro-nano structures after self-polishing, the polyacrylate-siloxane coatings are expected to demonstrate enhanced durability. Herein, static and dynamic immersion tests were carried out to evaluate the self-polishing rate of coatings as well as the durability of biomimetic surface. As shown in Fig. 4a and Fig. S10, the self-polishing ability of coatings was confirmed by the mass loss and thickness reduction after 3 days of static immersion. T21-N2, with no crosslinking networks, possessed the highest mass loss rate of 0.64 mg/cm^2 as well as the highest thickness reduction after ASW immersion, while K21-N2 possessed the lowest mass loss rate of 0.15 mg/cm^2 and showed no significant thickness reduction. The introduction of TISM could remarkably increase the mass loss rate and thickness reduction of cross-

linked polyacrylate-siloxane coatings, as T7K21-N2, T14K14-N2 and T21K7-N2 exhibited the mass loss rate of 0.30, 0.36 and 0.44 mg/cm^2 , respectively. The water absorption ratio of coatings was also measured during static immersion, as shown in Fig. S11. Similar to the pattern of the mass loss rate, T21-N2 exhibited the highest water absorption ratio of 14.4 %. The water absorption ratios of T21K7-N2, T14K14-N2, T7K21-N2 and K21-N2 significantly reduced to 7.6 %, 6.9 %, 5.0 % and 4.2 %, respectively, as a result of the blocking effect of crosslinking networks.

To further compare the self-polishing rate of coatings under different service conditions, the mass loss rate of coatings after static immersion and the one after dynamic immersion were presented in Fig. 4b. It could

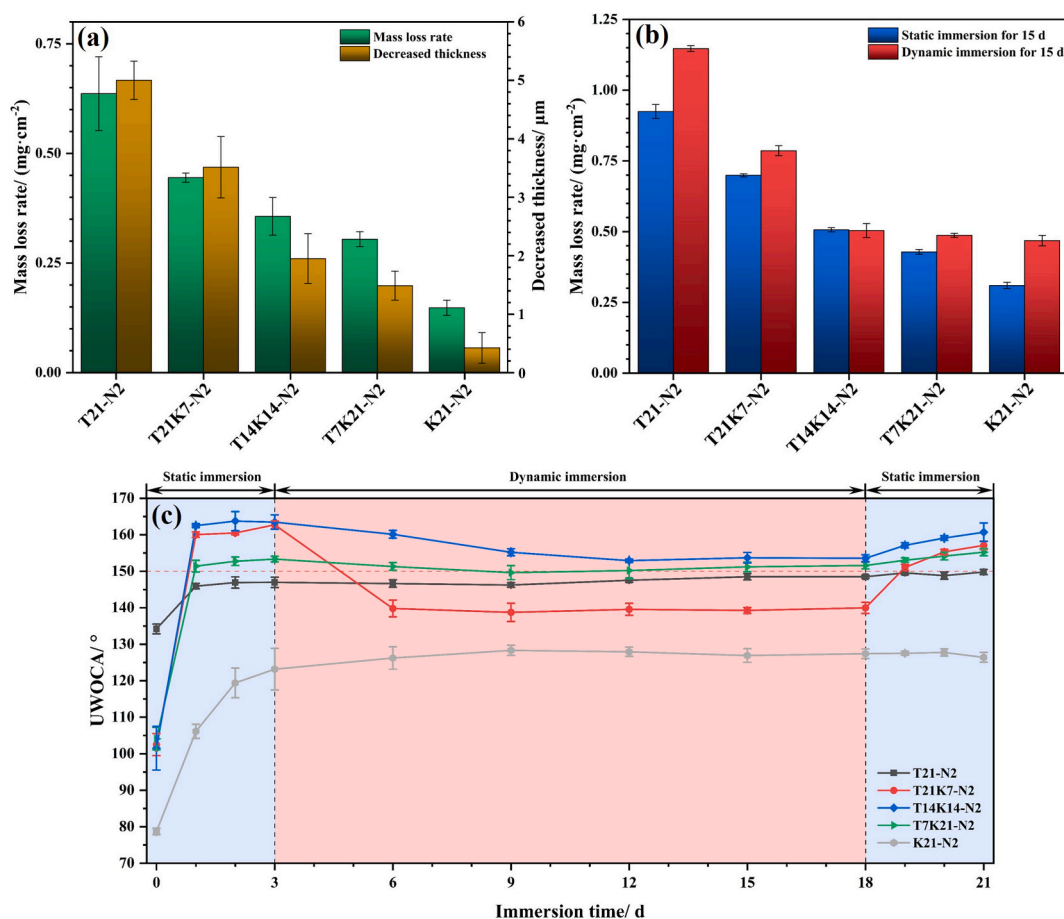


Fig. 4. (a) Thickness reduction and mass loss rate of polyacrylate-siloxane coatings after 3 days of static immersion; (b) the mass loss rate of coatings after 15 days of static immersion or after 15 days of dynamic immersion; and (c) the UWOCAs of coatings throughout the 21-day test cycle. (Mean $\pm$ SE, $n = 3$).

be found that, compared to static immersion, T21-N2 and K21-N2 possessed obviously higher mass loss rate under dynamic immersion, due to the shearing force of ASW. Interestingly, T14K14-N2 presented nearly the same self-polishing rate under both static and dynamic conditions. This result might be related to the micro-nano structures and the alkaline hydration layer formed on the surface of the coating. That is, under static immersion, the porous micro-nano structures assured the existence of a surface hydration layer with high alkalinity, which promoted the hydrolysis of the coating surface. And under dynamic condition, the micro-nano structures suffered from mechanical wear, which significantly reduced the ability of coating surface to maintain alkaline hydration layer. The reduction in surface alkalinity brought about a slower hydrolysis rate of the resin matrix itself, thus achieving a relatively stable self-polishing rate.

Fig. 4c presents the changes in UWOCAs of coatings during static and dynamic immersion tests. To simulate the whole process of the actual voyage, a test cycle consisting of 3 days of initial static immersion, 15 days of dynamic immersion and another 3 days of static immersion was designed. During the first static immersion stage, the UWOCAs of all coatings increased as a result of the dissolution of amino-siloxanes and the hydrolysis of triisopropylsilyl ester group. Compared to T21-N2 and K21-N2, the coatings containing both self-polishing and cross-linking monomers in polyacrylate chains were able to obtain underwater superoleophobicity after 3 days of immersion, as T14K14-N2 showed the highest UWOCA of 163.5° . This result could be explained by the lyophobic effect of self-generated micro-nano structures with relatively high roughness (Fig. S12), which enhanced the underwater oleophobic property of the surfaces. During the dynamic immersion simulating 10 knots of the ship's navigational speed, the coatings with both TISM and

TMSPM content presented varying degrees of UWOCA reduction, due to the accelerated loss of hydrolyzed polyacrylate chains from the surfaces, causing adverse changes in the surface hydration layer and micro-nano structures. The UWOCA reduction of the above coatings under dynamic condition decreased as the content of TMSPM in the resin increased. Meanwhile, despite the decrease in UWOCAs, T14K14-N2 still maintained the highest UWOCA of 153.6° after 15 days of dynamic immersion. The coatings were then statically immersed again to simulate the arrival of ships at the port. It could be found that T7K21-N2, T14K14-N2 and T21K7-N2 all exhibited the increase in UWOCAs and the recovery of underwater superoleophobicity after 3 days of immersion. None of the coatings showed significant peeling after completing the test cycle. SEM cross section images of T14K14-N2 after immersion tests were also obtained (Fig. S13), showing a stable layer of micro-nano structures roughly 1 to 2 μm thick on the coating surface under both static and dynamic conditions. The results proved that, by combining the seawater-induced self-polishing groups and hydrolyzable cross-linking networks, the polyacrylate-siloxane antifouling coatings could form a recoverable underwater superoleophobic surface, leading to better fouling-resistant performance and durability than the traditional self-polishing coatings.

In addition to the shear effect of seawater, the coating surface might also suffer from direct mechanical damage caused by the impact of sediment or fishes. As shown in Fig. S14, the regeneration of micro-nano structures was observed on the deeply abraded surface of T14K14-N2 after ASW immersion for 1 h, further verifying the recovery ability of the biomimetic surface.

3.4. Laboratory antibacterial and antifouling assessment

The antibacterial performance of coatings was obtained by in-vitro bacteriostatic test, with the sterilizing rate representing the proportion of bacteria killed after 24 h of contact with coating surfaces. Firstly, two widely distributed bacteria, *Escherichia coli* and *Shewanella loihica*, were chosen as targets to evaluate the bactericidal effect of the initial coating surfaces. As shown in Fig. 5a, e, the polyacrylate-siloxane coatings all exhibited a certain degree of bactericidal effect against *E. coli* and *S. loihica*, compared to the glass substrate. Further comparison showed that the sterilizing rates of coatings increased with the number of amino groups in the curing agents, which was consistent with the change of surface alkalinity of coatings. The sterilizing rates of T14K14-N2 against *E. coli* and *S. loihica* were 97.3 % and 98.4 %, respectively, while those of T14K14-N3 were 99.4 % and 99.6 %, proving that the rapid dissolution of amino-siloxanes in these coatings was able to create an uninhabitable surface for bacteria. Interestingly, although no significant surface alkalinity was detected, T14K14-N1 still exhibited the sterilizing rates of 52.5 % against *E. coli* and 46.6 % against *S. loihica*. This might be explained by the fact that the hydrolyzed groups on the coating surface increased the surface charges during immersion, which affected the structure of the bacterial cell membrane, leading to a certain bactericidal effect as well [35]. The bactericidal effect of alkaline hydration layer could be further verified through the time-dependent pH variation of the solution during in-vitro bacteriostatic test, as shown in Fig. 5b. The pH value of bacteria suspension in contact with glass control first decreased to about 5.7 as a result of the respiration of bacteria, and then increased to about 7.4 due to the protein putrefaction, indicating the massive reproduction of bacteria. In contrast, the pH of the solution on the surface of T14K14-N2 quickly increased to 8.5 after 1 h of immersion, while that of T14K14-N3 increased to 9.0. After that, neither T14K14-N2 nor T14K14-N3 showed significant variation in the pH of the solution during the testing, verifying the effective inhibition of bacterial growth by surface alkalinity of coatings.

The sterilizing rates of coatings against *E. coli* were measured again after 15 days of static and dynamic immersion to evaluate the long-term antibacterial performance of coatings. As shown in Fig. 5c, e, both T14K14-N2 and T14K14-N3 maintained the sterilizing rates of more than 90 % after static immersion, showing better antibacterial durability than T14K14-N1. The results indicated that the alkaline hydration layer promoted the continuous self-polishing of the coating surface, so that the amino-siloxanes inside the coating could be steadily released from the newly exposed surface to maintain the bactericidal effect of the surface. However, after dynamic immersion, only T14K14-N2 still maintained the sterilizing rate higher than 90 % against *E. coli*, while the sterilizing rate of T14K14-N3 decreased to 82.4 %. As presented in Fig. 5d, the surface Zeta potentials of coatings after static and dynamic immersion were compared with those after only one day of ASW immersion to reflect the evolution of surface alkalinity of coatings. T14K14-N1 possessed negative surface Zeta potential, which was in consistent with its weak surface alkalinity. Both T14K14-N2 and T14K14-N3 showed only a slight decrease in surface Zeta potential after 15 days of static immersion, showing the stable hydrolysis and dissolution of the amino-siloxanes under static condition. However, after 15 days of dynamic immersion, the Zeta potential on the surface of T14K14-N3 showed a significantly greater decrease than that of T14K14-N2. As discussed in Section 3.3, this phenomenon might also be related to the slowed hydrolysis rate of T14K14-N2 under dynamic conditions, which preserved the amino-siloxanes inside the coating from rapid loss. Overall, T14K14-N2 exhibited the best antibacterial performance among coatings cured with different amino-silane coupling agents. Its sterilizing rate was comparable to that of the biocide-modified biomimetic surfaces [28,29] and mechano-bactericidal nanopillar surfaces [36], much higher than that of the conventional micro-nano structures [37]. Moreover, thanks to the unique self-regeneration ability of biomimetic surfaces, the antibacterial durability of polyacrylate-siloxane coatings

was significantly better than that of the above surfaces.

The fouling-resistant ability of coatings was evaluated through the adhesion of proteins and algae on the surface. Fig. 6a, c shows the relative fluorescence intensity of FITC-BSA observed by a confocal laser scanning microscope, using glass substrate as control. It could be found that T21-N2 possessed better anti-protein property than the glass substrate, while the surface of K21-N2 exhibited the highest protein adhesion density, which demonstrated that a self-polishing surface with decent underwater oleophobicity was more likely to inhibit the attachment of protein molecules to its surface. As for those coatings based on trimethoxysilyl/triisopropylsilyl ester-bearing polyacrylate, T14K14-N2 carried the best anti-protein adhesion ability, with a relative fluorescence intensity of only 0.183 on its surface. This trend was in good agreement with the test results of UWOCAs. Fig. 6b, c shows the density of *Chlorella* and *Cyclotella cryptica* settled on the coating surface. On contrary to the protein adhesion, K21-N2 exhibited better anti-algae property than the glass substrate, while T21-N2 possessed the highest density of adhered microalgae on the surface. The density of microalgae settled on the surface of T7K21-N2 and T21K7-N2 was between that of K21-N2 and T21-N2. According to previous reports, the settlement of microalgae depends on the surface charge rather than on the wettability [38,39]. This is because most marine microalgae possess negative surface charge, which can form electrostatic interaction with the positively charged coating surface, leading to an increased adhesion to these surfaces. Therefore, with the increased content of self-polishing monomer and the decreased content of cross-linking monomer, the coating surface could absorb more water in a shorter time and therefore facilitated the hydrolysis and dissolution of the matrix, resulting in more protonated amino groups exposed on the surface (Fig. S15). However, T14K14-N2 still maintained the best anti-algae property among all coatings, presenting algal adhesion reduction of 85.1 % and 78.6 % against *Chlorella* and *Cyclotella cryptica* compared to the glass substrate. This result suggested that the micro-nano structures formed on the coating surfaces also had an influence on the anti-algae property of coatings (Fig. S12). Given that the size of the single-celled microalgae was normally between 1 and 10 μm , the micro-nano structure of T14K14-N2 provided a minimum contact area through the appropriate pore size and the highest surface porosity. Since the protonated amino groups mainly existed at the phase interface, the micro-nano porous structure could also prevent the microalgae from contacting the positively charged coating matrix beneath the structures. Thus, although T14K14-N2 exhibited high positive potential, the microalgae were unable to form firm interaction with its surface.

3.5. Marine field test

Marine field test was carried out at the offshore area of Yellow Sea during the booming season of marine organisms. Fig. 7 displays the photos of blank and coated plates before and after marine field test. After 15 days of static submerging test, the blank plate exhibited the severest biofouling on its surface, as the green algae were firmly settled and grew vigorously. The coatings containing only one functional monomer (T21-N2, K21-N2) and the coatings with low self-polishing rates (T14K14-N1, T7K21-N2) also exhibited surfaces completely covered by fouling organisms. Among all coatings, T14K14-N2 showed the best antifouling performance under static condition, with minimum covered area of biofouling and no mature green algae observed on its surface. As for the plates after 50 days of dynamic test, the settlement of macro fouling organisms on the surfaces was significantly less than that of static test, as a result of the shear force of seawater. The surfaces of T21-N2, K21-N2, T14K14-N1 and the blank plate were contaminated with a layer of brown slime, which might be related to the adhesion of microorganisms and the formation of biofilms. In comparison, the attachment of brown slime was greatly reduced on the surfaces of T14K14-N2, T14K14-N3, T21K7-N2 and T7K21-N2. The results verified that T14K14-N2 could maintain the outstanding antifouling performance under

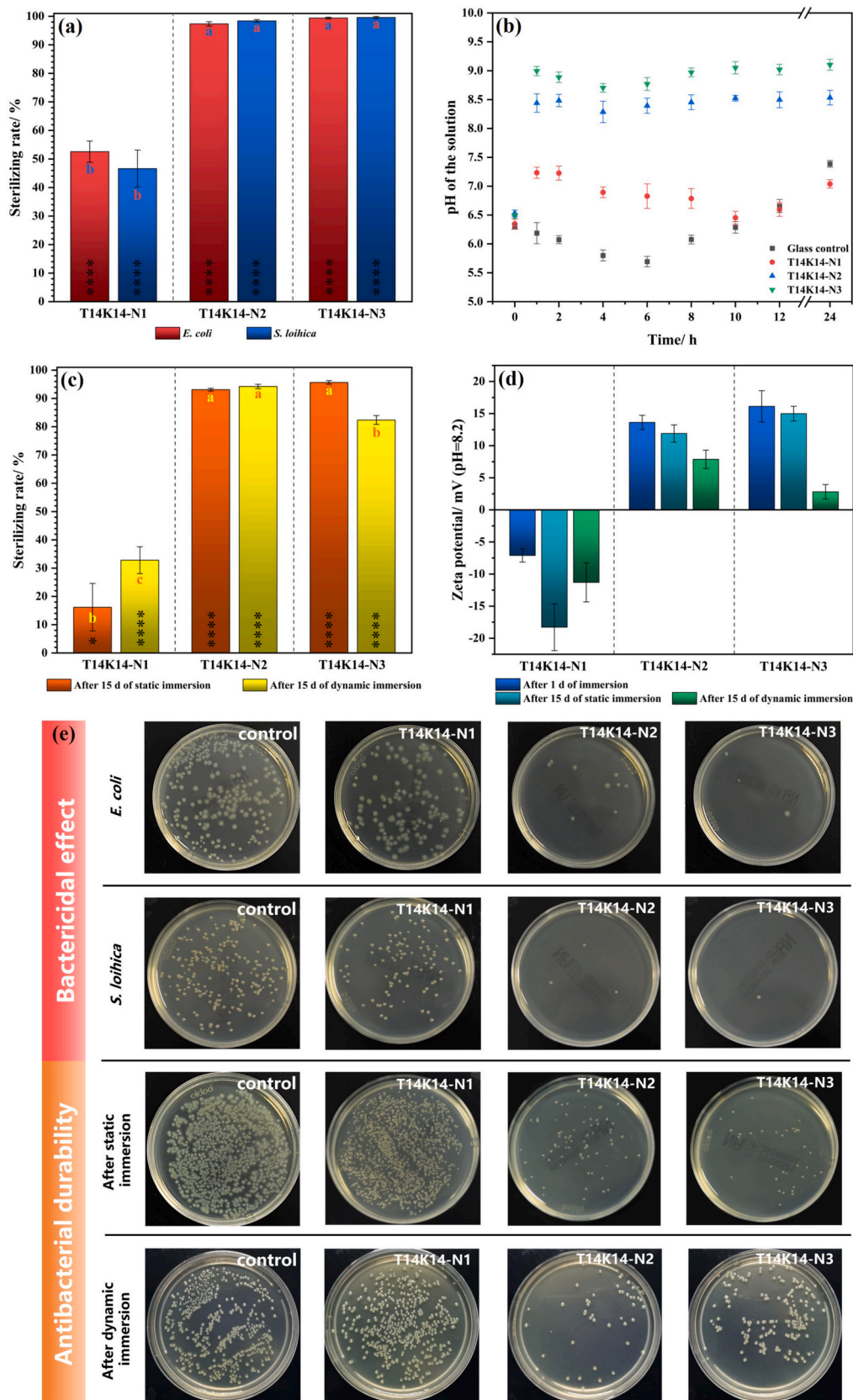


Fig. 5. (a) Sterilizing rate of coatings against *E. coli* and *S. loihica*; (b) time-dependent pH variation of the solution during in-vitro bacteriostatic test against *E. coli*; (c) antibacterial durability of coatings against *E. coli* after 15 days of static and dynamic immersion; (d) the surface Zeta potentials of different coatings after static and dynamic immersion; (e) agar plate photographs of in-vitro bacteriostatic tests. (Mean $\pm$ SE, $n = 3$; asterisks at the bottom of bars indicate statistically significant differences vs. the glass control: * $p < 0.05$ and **** $p < 0.0001$ by one-way ANOVA; groups that are significantly different between each other at $p < 0.05$ are indicated by the different letters below the error bars; the raw data of in-vitro bacteriostatic test are listed in Table S2.).

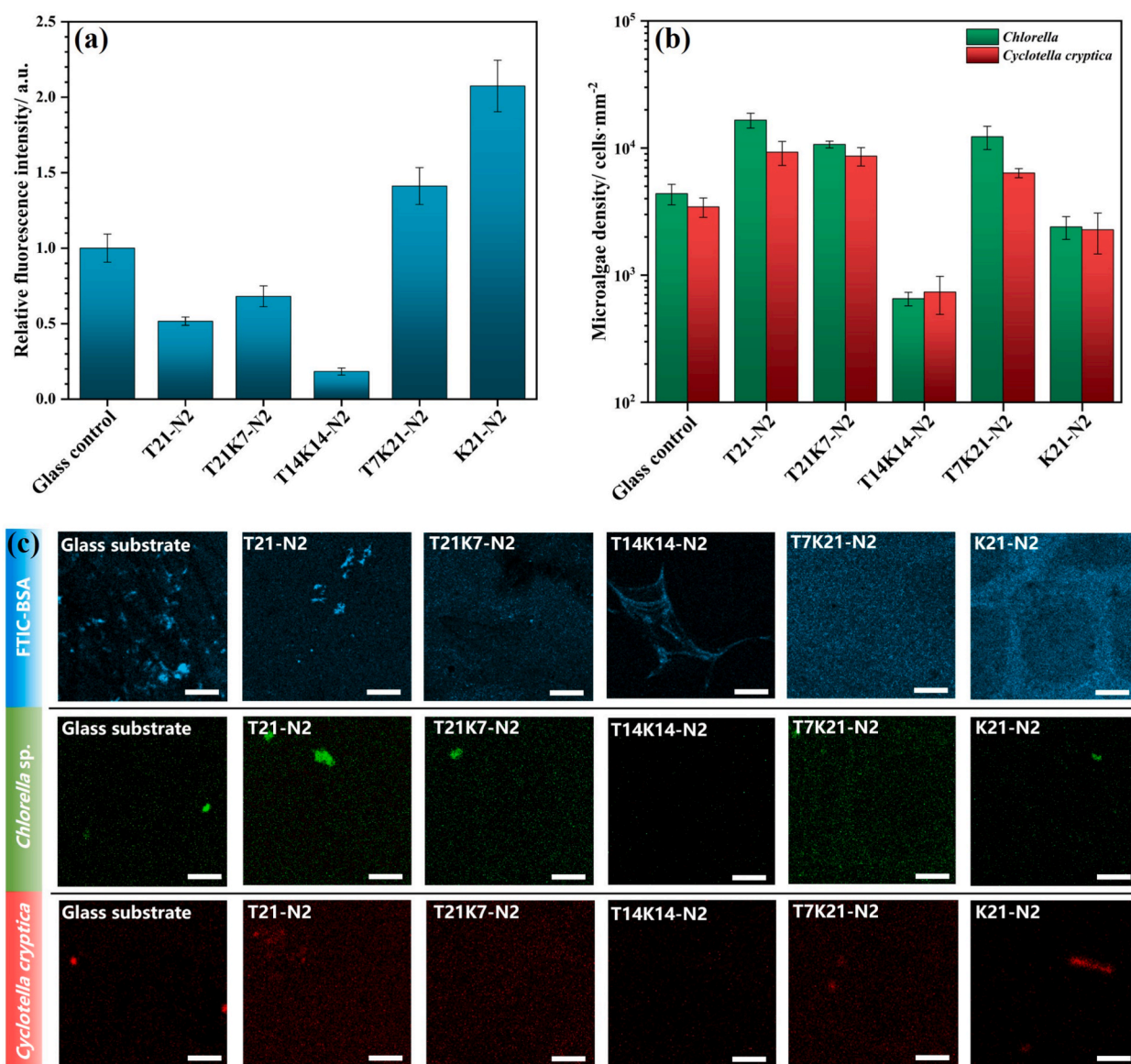


Fig. 6. (a) Relative fluorescence intensity of FITC-BSA on the coating surface; (b) *Chlorella* and *Cyclotella cryptica* density on the coating surface; and (c) CLSM images of laboratory antifouling tests. (Mean $\pm$ SE, $n = 3$; scale bar = 100 μm .)

varied conditions, with the ability to inhibit the adhesion of both micro and macro fouling organisms. No significant biofoulings were observed on its surface after 15 days of static test or after 50 days of dynamic test, which was able to meet the needs of most ships during regular sailing and docking process.

It is known that the skins of some marine mammals, such as pilot whales and dolphins, show excellent antifouling performance against algae and barnacles. These skins are reported to share a similar antifouling mechanism, which can be summarized as the combination of surface micro-nano structures, a layer of bactericidal secretions surrounding the structures, and the ability of continuous peeling and regeneration of the skin [23,40–42]. As illustrated in Scheme 2, the antifouling effect of the polyacrylate-siloxane coatings can be explained by the biomimetic structures and functions that generally match the skins of marine mammals. According to previous discussion, the polyacrylate-siloxane coatings could spontaneously separate into a dispersed phase enriched with amino-siloxanes and a continuous phase enriched with polyacrylate. During seawater immersion, the dispersed phase was quickly hydrolyzed and dissolved from the surface, while the continuous phase relied on the gradual hydrolysis of silyl ester and

siloxane groups to achieve the underwater superoleophobicity of the surface. The above process led to the formation of an ordered biomimetic micro-nano structure on the coating surface, which, under appropriate roughness, could reduce the contact and the adhesion of microorganisms to the surface. Meanwhile, the rapid dissolution of amino-siloxanes also resulted in a surface hydration layer with strong alkalinity, which provided an antibacterial barrier around the surface micro-nano structures like the skins of pilot whales and dolphins. Additionally, under the combined action of the micro-nano structures and the alkaline hydration layer, the coating such as T14K14-N2 could maintain a stable and adequate self-polishing rate of its surface under different conditions, improving the antifouling durability in the actual marine environment.

4. Conclusions

Bactericide-free polyacrylate-siloxane coatings were easily prepared with the polyacrylate bearing trimethoxysilyl/triisopropylsilyl ester pendant groups and amino-silane coupling agent. When the amino-silane coupling agent contained more than one amino group, the

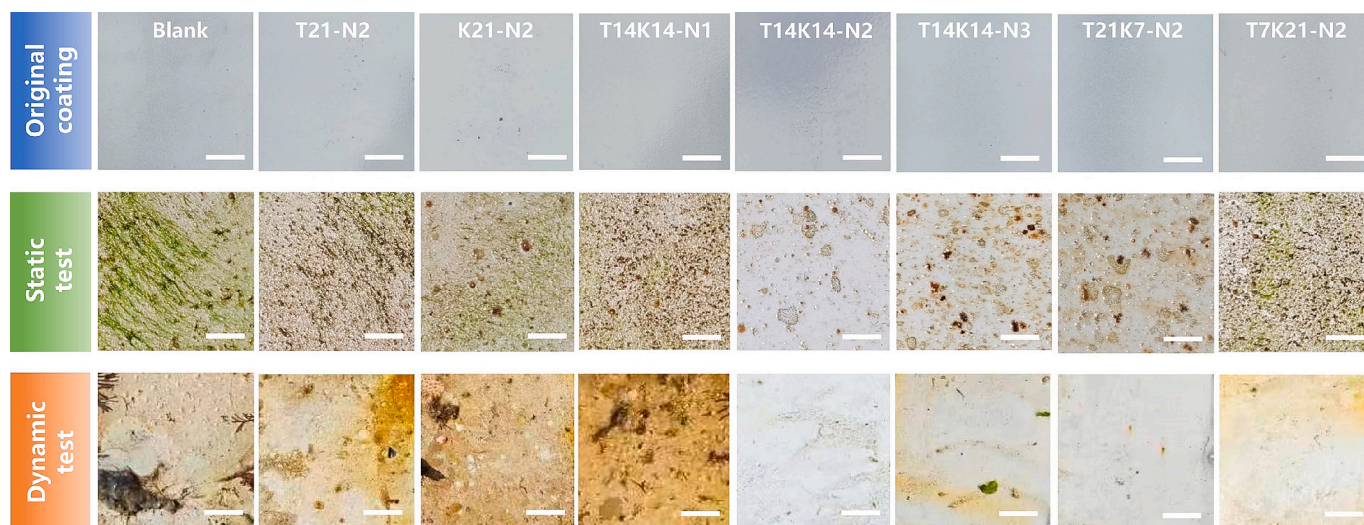
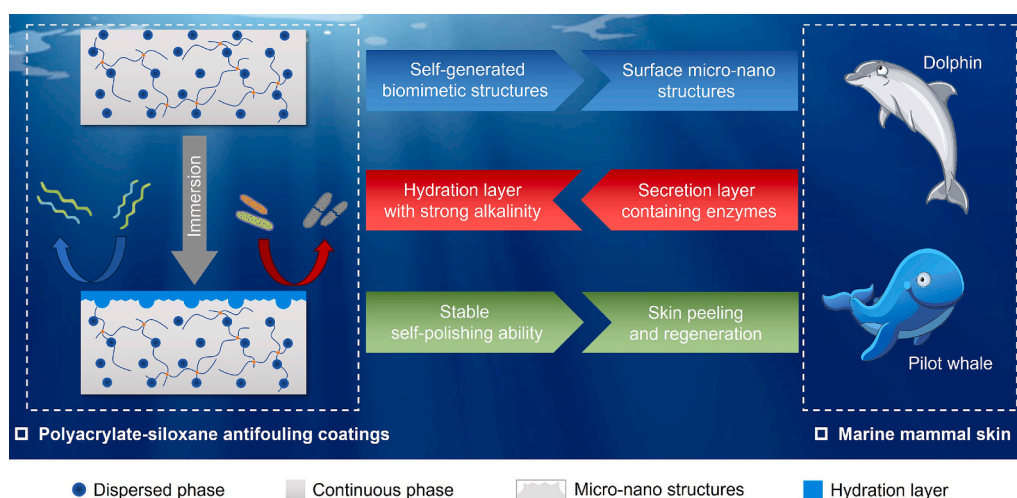


Fig. 7. Digital photos of the testing plates before and after marine field test. (Scale bar = 2 cm.).



Scheme 2. Schematic diagram of biomimetic antifouling mechanism of the polyacrylate-siloxane coatings.

coating could spontaneously separate into a dispersed phase enriched with amino-siloxane and a continuous phase enriched with polyacrylate, which could self-generate a biomimetic surface with micro-nano structure and an alkaline hydration layer after seawater immersion. Interestingly, the coating, i.e., T14K14-N2, based on the polyacrylate containing 1:1 ratio of TISM and TMSPM content and [3-(2-aminoethyl) aminopropyl]trimethoxysilane exhibited identical self-polishing rate under both static and dynamic conditions. This coating not only possessed the ability to effectively deal with different biofoulings in the laboratory antifouling assessment, including the sterilizing rate of 97.3 %, anti-protein rate of 81.7 % and anti-algae rate of 85.1 %, but also exerted outstanding antifouling performance under varied conditions of the actual marine environment. Overall, the antifouling effect of the polyacrylate-siloxane coatings could be explained by the combination of fouling-resistant micro-nano structures, bactericidal hydration layer and stable self-polishing ability, which generally matched the skins of some marine mammals. By virtue of the facile and economical fabrication method, our work would provide a practical strategy to develop biomimetic antifouling coatings with promising application in the actual marine environment.

CRediT authorship contribution statement

Haofeng Xiao: Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Kai Zhang:** Methodology, Formal analysis. **Taijiang Gui:** Writing – review & editing, Supervision. **Shuxue Zhou:** Writing – review & editing, Supervision, Resources, Project administration, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.cej.2025.164087>.

Data availability

Data will be made available on request.

References

- [1] L. Li, H. Hong, J. Cao, Y. Yang, Progress in marine antifouling coatings: current status and prospects, *Coatings* 13 (2023) 1893, <https://doi.org/10.3390/coatings13111893>.
- [2] L.D. Chambers, K.R. Stokes, F.C. Walsh, R.J.K. Wood, Modern approaches to marine antifouling coatings, *Surf. Coat. Technol.* 201 (2006) 3642–3652, <https://doi.org/10.1016/j.surfcoat.2006.08.129>.
- [3] M.S. Selim, M.A. Shenashen, S.A. El-Safty, S.A. Higazy, M.M. Selim, H. Isago, A. Elmarakbi, Recent progress in marine foul-release polymeric nanocomposite coatings, *Prog. Mater. Sci.* 87 (2017) 1–32, <https://doi.org/10.1016/j.pmatsci.2017.02.001>.
- [4] C.A. Del Grosso, C. Leng, K. Zhang, H. Hung, S. Jiang, Z. Chen, J.J. Wilker, Surface hydration for antifouling and bio-adhesion, *Chem. Sci.* 11 (2020) 10367–10377, <https://doi.org/10.1039/D0SC03690K>.
- [5] S. Wang, B. Qiu, J. Shi, M. Wang, Quaternary ammonium antimicrobial agents and their application in antifouling coatings: a review, *J. Coat. Technol. Res.* 21 (2024) 87–103, <https://doi.org/10.1007/s11998-023-00825-z>.
- [6] P. Hu, Q. Xie, C. Ma, G. Zhang, Silicone-based fouling-release coatings for marine antifouling, *Langmuir* 36 (2020) 2170–2183, <https://doi.org/10.1021/acs.langmuir.9b03926>.
- [7] C. Bressy, A. Margailan, Erosion study of poly(trialkylsilyl methacrylate)-based antifouling coatings, *Prog. Org. Coat.* 66 (2009) 400–405, <https://doi.org/10.1016/j.porgcoat.2009.09.003>.
- [8] T. Ding, L. Xu, X. Liu, L. Ma, Y. Cui, D. Li, X. Sun, C. Gao, A Cu₂O-based marine antifouling coating with controlled release of copper ion mediated by amphiphilic PLMA-b-PDMAEMA copolymers, *Prog. Org. Coat.* 170 (2022) 107003, <https://doi.org/10.1016/j.porgcoat.2022.107003>.
- [9] M. Lejars, A. Margailan, C. Bressy, Siloxy silyl ester methacrylate diblock copolymer-based coatings with tunable erosion and marine antifouling properties, *ACS Appl. Polym. Mater.* 2 (2020) 3291–3300, <https://doi.org/10.1021/acscpm.0c00434>.
- [10] Y. Liu, X. Suo, Z. Wang, Y. Gong, X. Wang, H. Li, Developing polyimide-copper antifouling coatings with capsule structures for sustainable release of copper, *Mater. Des.* 130 (2017) 285–293, <https://doi.org/10.1016/j.matdes.2017.05.075>.
- [11] X. Zang, Y. Ni, Q. Wang, Y. Cheng, J. Huang, X. Cao, C.J. Carmalt, Y. Lai, D. Ha Kim, Y. Liu, Z. Lin, Non-toxic evolution: Advances in multifunctional antifouling coatings, *Mater. Today* 75 (2024) 210–243, <https://doi.org/10.1016/j.mattod.2024.03.018>.
- [12] F. Song, J. Wang, L. Zhang, R. Chen, Q. Liu, J. Liu, J. Yu, P. Liu, J. Duan, Synergistically improved antifouling efficiency of a bioinspired self-renewing interface via a borneol/ boron acrylate polymer, *J. Colloid. Interface. Sci.* 612 (2022) 459–466, <https://doi.org/10.1016/j.jcis.2021.12.187>.
- [13] Y. Wang, C. Ni, H. Xu, Q. Tian, G. Song, G. Chen, X. Li, L. Yu, X. Yan, Preparation of acrylic metal salt resin containing capsaicin derivative structure and study of antifouling properties, *Prog. Org. Coat.* 186 (2024) 108085, <https://doi.org/10.1016/j.porgcoat.2023.108085>.
- [14] J. Sha, X. Liu, R. Chen, J. Yu, Q. Liu, J. Liu, J. Zhu, P. Liu, R. Li, J. Wang, Surface hydrolysis-anchored eugenol self-polishing marine antifouling coating, *J. Colloid. Interface. Sci.* 637 (2023) 67–75, <https://doi.org/10.1016/j.jcis.2023.01.058>.
- [15] X. Ai, L. Mei, C. Ma, G. Zhang, Degradable hyperbranched polymer with fouling resistance for antifouling coatings, *Prog. Org. Coat.* 153 (2021) 106141, <https://doi.org/10.1016/j.porgcoat.2021.106141>.
- [16] C. Ma, W. Zhang, G. Zhang, P. Qian, Environmentally friendly antifouling coatings based on biodegradable polymer and natural antifoulant, *ACS Sustain. Chem. Eng.* 5 (2017) 6304–6309, <https://doi.org/10.1021/acssuschemeng.7b01385>.
- [17] W. Xu, C. Ma, J. Ma, T. Gan, G. Zhang, Marine biofouling resistance of polyurethane with biodegradation and hydrolyzation, *ACS Appl. Mater. Interfaces* 6 (2014) 4017–4024, <https://doi.org/10.1021/am4054578>.
- [18] A.M.C. Maan, A.H. Hofman, W.M. de Vos, M. Kamperman, Recent developments and practical feasibility of polymer-based antifouling coatings, *Adv. Funct. Mater.* 30 (2020) 2000936, <https://doi.org/10.1002/adfm.202000936>.
- [19] V.B. Damodaran, N.S. Murthy, Bio-inspired strategies for designing antifouling biomaterials, *Biomater. Res.* 20 (2016) 18, <https://doi.org/10.1186/s40824-016-0064-4>.
- [20] A. Gao, Y. Yan, T. Li, F. Liu, Biomimetic urchin-like surface based on poly (lactic acid) membrane for robust anti-wetting and anti-bacteria properties, *Mater. Lett.* 237 (2019) 240–244, <https://doi.org/10.1016/j.matlet.2018.11.063>.
- [21] Y. Liu, G. Li, A new method for producing "Lotus Effect" on a biomimetic shark skin, *J. Colloid. Interface. Sci.* 388 (2012) 235–242, <https://doi.org/10.1016/j.jcis.2012.08.033>.
- [22] A.M. Brzozowska, F.J. Parra-Velandia, R. Quintana, Z. Xiaoying, S.S.C. Lee, L. Chin-Sing, D. Jańczewski, S.L.M. Teo, J.G. Vancso, Biomimicking micropatterned surfaces and their effect on marine biofouling, *Langmuir* 30 (2014) 9165–9175, <https://doi.org/10.1021/la502006s>.
- [23] X. Cao, M.E. Pettitt, F. Wode, M.P. Arpa Sancer, J. Fu, J. Ji, M.E. Callow, J. A. Callow, A. Rosenhahn, M. Grunze, Interaction of zoospores of the green alga *Ulva* with bioinspired micro- and nanostructured surfaces prepared by polyelectrolyte layer-by-layer self-assembly, *Adv. Funct. Mater.* 20 (2010) 1984–1993, <https://doi.org/10.1002/adfm.201000242>.
- [24] Y. Xu, X. Luan, P. He, D. Zhu, R. Mu, Y. Wang, G. Wei, Fabrication and functional regulation of biomimetic interfaces and their antifouling and antibacterial applications: a review, *Small* 20 (2024) 2308091, <https://doi.org/10.1002/smll.202308091>.
- [25] L. Chen, Y. Duan, M. Cui, R. Huang, R. Su, W. Qi, Z. He, Biomimetic surface coatings for marine antifouling: Natural antifoulants, synthetic polymers and surface microtopography, *Sci. Total Environ.* 766 (2021) 144469, <https://doi.org/10.1016/j.scitotenv.2020.144469>.
- [26] M. Munther, T. Palma, I.A. Angeron, S. Salari, H. Ghassemi, M. Vasefi, A. Beheshti, K. Davami, Microfabricated biomimetic placoid scale-Inspired surfaces for antifouling applications, *Appl. Surf. Sci.* 453 (2018) 166–172, <https://doi.org/10.1016/j.apsusc.2018.05.030>.
- [27] Z. Tong, S. Zhang, S. Chen, R. Sun, P. He, L. Song, J. Hu, Y. Hou, X. Zhan, Q. Zhang, Sea anemone-inspired slippery liquid-infused porous surface (SLIPS) with bionic cilia for responsive 4D antifouling, *Small* 20 (2024) 2401658, <https://doi.org/10.1002/smll.202401658>.
- [28] D. Wang, J. Xu, J. Yang, S. Zhou, Preparation and synergistic antifouling effect of self-renewable coatings containing quaternary ammonium-functionalized SiO₂ nanoparticles, *J. Colloid. Interface. Sci.* 563 (2020) 261–271, <https://doi.org/10.1016/j.jcis.2019.12.086>.
- [29] L. Zhao, R. Chen, J. Liu, Q. Liu, J. Yu, J. Zhu, P. Liu, J. Wang, Layer-by-layer-assembled biomimetic microstructure surface with multiple synergistic antifouling performance, *Prog. Org. Coat.* 183 (2023) 107812, <https://doi.org/10.1016/j.porgcoat.2023.107812>.
- [30] A.V. Bers, E.R. Díaz, B.A.P. Da Gama, F. Vieira-Silva, S. Dobretsov, N. Valdivia, M. Thiel, A.J. Scardino, C.D. McQuaid, H.E. Sudgen, J.C. Thomason, M. Wahl, Relevance of mytilid shell microtopographies for fouling defence - a global comparison, *Biofouling* 26 (2010) 367–377, <https://doi.org/10.1080/08927011003605888>.
- [31] C.M. Magin, S.P. Cooper, A.B. Brennan, Non-toxic antifouling strategies, *Mater. Today* 13 (2010) 36–44, [https://doi.org/10.1016/S1369-7021\(10\)70058-4](https://doi.org/10.1016/S1369-7021(10)70058-4).
- [32] X. Zhang, W. Zhang, J. Jing, L. Zhu, H. Ning, H. Liu, M. Li, H. Jin, H. Wang, H. Wang, Z. Liu, Combination of chemical anchoring and microcapsules for durable slippery coating with anti-corrosion/scaling property, *Prog. Org. Coat.* 200 (2025) 108992, <https://doi.org/10.1016/j.porgcoat.2024.108992>.
- [33] Z. Xiong, J. Liu, Y. Yang, Q. Lai, X. Wu, J. Yang, Q. Zeng, G. Zhang, S. Zhao, Reinforcing hydration layer on membrane surface via nano-capturing and hydrothermal crosslinking for fouling reduction, *J. Membr. Sci.* 644 (2022) 120076, <https://doi.org/10.1016/j.memsci.2021.120076>.
- [34] H. Yan, Q. Wu, C. Yu, T. Zhao, M. Liu, Recent progress of biomimetic antifouling surfaces in marine, *Adv. Mater. Interfaces* 7 (2020) 2000966, <https://doi.org/10.1002/admi.202000966>.
- [35] J. Chen, T. Qin, Y. Cheng, C. Fang, X. Zhang, M. Chen, Surface charge regulation of imidazolium salts/starch/carboxymethyl cellulose ternary polymer blend films for controlling antimicrobial performance and hydrophobicity, *Int. J. Biol. Macromol.* 245 (2023) 125519, <https://doi.org/10.1016/j.jbiomac.2023.125519>.
- [36] S.M. Kelleher, O. Habimana, J. Lawler, B.O. Reilly, S. Daniels, E. Casey, A. Cowley, Cicada wing surface topography: an investigation into the bactericidal properties of nanostructural features, *ACS Appl. Mater. Interfaces* 8 (2016) 14966–14974, <https://doi.org/10.1021/acscami.5b08309>.
- [37] Y. Chang, E.R. Weeks, W.A. Ducker, Surface topography hinders bacterial surface motility, *ACS Appl. Mater. Interfaces* 10 (2018) 9225–9234, <https://doi.org/10.1021/acscami.7b16715>.
- [38] J. Do, S.M. Kang, Effect of surface charges in polymer coatings on antifouling performance in marine environments with sediment, *Langmuir* 40 (2024) 19644–19653, <https://doi.org/10.1021/acs.langmuir.4c02304>.
- [39] E. Evariste, C.M. Gatlery, M.R. Detty, M.E. Callow, J.A. Callow, The performance of aminoalkyl/fluorocarbon/hydrocarbon-modified xerogel coatings against the marine alga *Ectocarpus cruaniorum*: relative roles of surface energy and charge, *Biofouling* 29 (2013) 171–184, <https://doi.org/10.1080/08927014.2012.758717>.
- [40] C. Baum, W. Meyer, R. Stelzer, L.G. Fleischer, D. Siebers, Average nanorough skin surface of the pilot whale (*Globicephala melas*, Delphinidae): considerations on the self-cleaning abilities based on nanoroughness, *Mar. Biol.* 140 (2002) 653–657, <https://doi.org/10.1007/s00227-001-0710-8>.
- [41] Y. Qin, S. Wang, Y. Fan, L. Wang, C. Zhang, J. Zhao, L. Ren, Osthole-infused polyurethane flexible coatings for enhanced underwater drag reduction and robust anti-biofouling, *Prog. Org. Coat.* 188 (2024) 108213, <https://doi.org/10.1016/j.porgcoat.2024.108213>.
- [42] D.K. Wainwright, F.E. Fish, S. Ingersoll, T.M. Williams, J. St Leger, A.J. Smits, G. V. Lauder, How smooth is a dolphin? The ridged skin of odontocetes, *Biol. Lett.* 15 (2019) 20190103, <https://doi.org/10.1098/rsbl.2019.0103>.